Webcam Full-Body Retargeting for MABEL: Anatomical Hand Decomposition, Depth Restoration, and Two-Hand Consistency

MABEL simulation control center — retargeting subsystem

July 2, 2026

Abstract

The desktop webcam teleoperation path synthesizes Vision-Pro-style `teleop_frame` messages (head 4×4, per-hand anchor + 25 named ARKit joint transforms) from MediaPipe PoseLandmarker world landmarks and HandLandmarker hand landmarks, so the unchanged server-side retargeting engine (finger keypoint IK, cosine-directional arm IK, neck, whole-body followers) drives the robot. This document derives and validates a rigorous upgrade of that frame synthesis: (i) an *anatomical hand model* — the 21 landmarks are decomposed into per-joint angles (abduction in the palm plane, MCP/PIP/DIP flexion about per-finger axes, a thumb direction + MP/IP chain) and re-synthesized with calibrated bone lengths, with all temporal filtering in the *angle* domain; (ii) *depth restoration* — monocular hand landmarks systematically attenuate depth, which is corrected by a per-hand global attenuation estimate plus per-bone rigid-length restoration anchored by anthropometric priors; (iii) *two-hand consistency* — one similarity transform maps the whole operator (wrist placement *and* finger geometry), and a shared-image-frame fusion corrects the inter-wrist vector, so fingertips that touch across the operator's hands touch on the wire exactly; (iv) *elbow streaming* — the webcam tracks elbows (a headset does not), and a minimal server addition feeds them to the arm IK's existing elbow-direction objective. Validation replays a scripted landmark corpus (twelve canonical shapes with exact ground truth) through the live simulation stack. On ideal landmarks the upgraded synthesis is *exact* (fingertip-gap and contact errors = 0; legacy: 18 mm to 29 mm); under a MediaPipe-degradation model the fist pitch error drops from 13.5° to 8.7° RMSE, pitch step-response lag drops from 86 ms to 19 ms, and at the robot after physics the pinch fingertip gap error drops 61 mm→24 mm and the two-hand touch gap 131 mm→97 mm. The harness also uncovered (and compensates client-side) a chirality inversion in the simulated ORCA hand's finger-curl convention, which was making curling fingers *extend*.

Keywords: teleoperation, retargeting, MediaPipe, hand pose, depth ambiguity, bimanual coordination, one-euro filter, MABEL.

1 Overview and the server contract

1.1 Pipeline

The browser client (`simulation_control_center/web/js/bodyteleop.js`) runs MediaPipe `tasks-vision` on the webcam at ∼30 Hz: PoseLandmarker yields 33 *world* landmarks (metric, body-centered) plus normalized image landmarks; HandLandmarker yields, per hand, 21 *world* landmarks (metric, hand-centered) and 21 normalized image landmarks. From these the client synthesizes the same wire frames the Vision Pro app streams;

the WebSocket bridge (`server/sim_teleop_bridge.py`) feeds them through `mabel.wire` → `OperatorPose` → the `TeleopEngine` (`simulation/mabel_mujoco/scripts/teleop/engine.py`) composed by the `MabelTeleop` facade. The engine's mathematics (finger cosine+relative SLSQP, cosine-directional arm IK, neck, head-follow whole body) is documented in `controller/docs/mabel_controller.tex` §3–§8 and is *not* duplicated here; this document covers the client-side synthesis and the validation of the pair.

1.2 Exactly what the server consumes

Auditing `mabel/wire.py`, `teleop/engine.py`, `teleop/hand.py`, `teleop/geometry.py`:

consumer	joints used	quantity
`wire.py` pinch/clutch	`wrist`, `thumb_tip`, `index_tip`	thumb–index *distance* [m]
`geometry.operator_frame`	`wrist`, {index,middle,little}`_metacarpal`	palm frame R_{op} (orientation)
`engine` arm contact	`middle_tip`	absolute position
`engine` arm elbow (new)	`leftElbow/rightElbow`	direction from shoulder
`hand.py` finger IK	thumb 1–4; per finger knuckle, `intermediate_base`, tip	wrist-relative positions, rotated into R_{op}, metric

Finger-DIP landmarks (MediaPipe 7, 11, 15, 19) are carried but not read by the finger cost; they still shape the tip through the synthesis chain. The four metacarpals absent from MediaPipe are synthesized at fraction 0.35 of the wrist→knuckle segment.

2 Frames and the operator similarity transform

2.1 Camera frame to virtual ARKit frame

MediaPipe world coordinates have x = image right, y = image down, z = away from the camera (meters). The virtual ARKit operator frame has X = operator's right, Y = up, $-Z$ = operator's forward (toward the camera). Because the operator *faces* the camera,

$$\mathbf{p}_{\text{a}} = M\,\mathbf{p}_{\text{mp}}, \qquad M = \operatorname{diag}(-1,-1,1), \qquad \det M = +1, \tag{1}$$

a proper rotation (180° about z), so hand chirality is preserved exactly — no mirroring is introduced by the frame change itself (cf. §5.4).

2.2 Anchor, anthropometric scale, and the similarity transform

On engage (and on `recalibrate`) the shoulder-center world position $\mathbf{a}$ is captured together with the measured biacromial width w; the scale is $s = \text{clip}(0.38/w,\ 0.5,\ 3.0)$ (0.38 m nominal). Every operator point is mapped by *one similarity transform*

$$\mathcal{S}(\mathbf{p}) \;=\; s\,M(\mathbf{p}-\mathbf{a}) \;+\; h\,\hat{\mathbf{y}}, \qquad h = 1.35\,\text{m}, \tag{2}$$

which places a virtual standing operator of nominal size at the origin. The head transform (ear-midpoint position, gaze from nose/ears, $-Z$ gaze convention) is unchanged from the original implementation and validated against the server's neck math in the unit tests.

The upgrade applies $\mathcal{S}$ *uniformly*: previously the hand shape (hand-centered landmarks) was attached to the mapped wrist *unscaled*, which breaks cross-hand contact (§5).

3 Anatomical hand decomposition and re-synthesis

Raw MediaPipe hand landmarks carry (a) per-frame bone-length noise, (b) systematically attenuated depth, and (c) no anatomical consistency guarantee. Passing them through verbatim makes the server's shape-matching IK chase noise and under-curled geometry. Instead the client fits an anatomical model.

3.1 Palm frame

Let $\mathbf{p}_0$ be the wrist and $\mathbf{p}_5, \mathbf{p}_9, \mathbf{p}_{17}$ the index, middle and little knuckles of the wrist-centered shape. With the chirality factor $\chi = +1$ (right) / -1 (left):

$$\hat{\mathbf{y}} \propto \mathbf{p}_9 - \mathbf{p}_0, \qquad \hat{\mathbf{z}} \propto \chi\,(\mathbf{p}_5 - \mathbf{p}_0) \times (\mathbf{p}_{17} - \mathbf{p}_0) \text{ (orthogonalized against } \hat{\mathbf{y}}), \qquad \hat{\mathbf{x}} = \hat{\mathbf{y}} \times \hat{\mathbf{z}}. \tag{3}$$

Lemma (dorsal normal). For any true right-hand skeleton, in any orientation, $(\mathbf{p}_5 - \mathbf{p}_0) \times (\mathbf{p}_{17} - \mathbf{p}_0)$ points to the *dorsal* (back-of-hand) side; for a left hand it points ventral. Hence $\hat{\mathbf{z}}$ in (3) is always dorsal and fingers always flex toward $-\hat{\mathbf{z}}$. This is a chirality invariant — it cannot be changed by any proper rotation of the hand — and is the basis of both the decomposition below and the convention measurement in §5.4.

3.2 Finger decomposition

For finger f with knuckle $\mathbf{k}$ and phalanx vectors $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$ (proximal, middle, distal):

$$\hat{\mathbf{m}} \propto \mathbf{k} - \hat{\mathbf{z}}(\hat{\mathbf{z}}\cdot\mathbf{k}) \qquad \text{(palm-plane metacarpal ray)}, \tag{4}$$

$$\varphi = w_\varphi \operatorname{atan2}\big((\hat{\mathbf{m}} \times \hat{\mathbf{d}}_1^{\parallel})\cdot\hat{\mathbf{z}},\ \hat{\mathbf{m}}\cdot\hat{\mathbf{d}}_1^{\parallel}\big), \quad w_\varphi = \operatorname{clip}\Big(\tfrac{\|\mathbf{d}_1^{\parallel}\|}{0.35\|\mathbf{d}_1\|}, 0, 1\Big) \qquad \text{(abduction)}, \tag{5}$$

$$\hat{\mathbf{r}} = R(\hat{\mathbf{z}}, \varphi)\,\hat{\mathbf{m}}, \qquad \hat{\mathbf{a}} = \hat{\mathbf{z}} \times \hat{\mathbf{r}} \qquad \text{(flexion axis)}, \tag{6}$$

$$\theta_{\text{mcp}} = \angle_{\hat{\mathbf{a}}}(\hat{\mathbf{r}} \to \hat{\mathbf{d}}_1), \quad \theta_{\text{pip}} = \angle_{\hat{\mathbf{a}}}(\hat{\mathbf{d}}_1 \to \hat{\mathbf{d}}_2), \quad \theta_{\text{dip}} = \angle_{\hat{\mathbf{a}}}(\hat{\mathbf{d}}_2 \to \hat{\mathbf{d}}_3), \tag{7}$$

where $\mathbf{d}_1^{\parallel}$ is the palm-plane projection of $\mathbf{d}_1$, $\angle_{\hat{\mathbf{a}}}(\mathbf{u} \to \mathbf{v}) = \operatorname{atan2}((\mathbf{u} \times \mathbf{v})\cdot\hat{\mathbf{a}}, \mathbf{u}\cdot\mathbf{v})$ is the signed angle about the flexion axis, and $R(\hat{\mathbf{a}}, \theta)$ is the Rodrigues rotation. Signs are chosen so $\theta > 0$ curls toward the palm ($-\hat{\mathbf{z}}$): rotating $\hat{\mathbf{r}}$ about $\hat{\mathbf{a}} = \hat{\mathbf{z}} \times \hat{\mathbf{r}}$ gives $\hat{\mathbf{d}}_1 = \hat{\mathbf{r}}\cos\theta - \hat{\mathbf{z}}\sin\theta$. The abduction fade w_φ removes the spherical-coordinate singularity at $\theta_{\text{mcp}} = 90°$: the palm-plane projection of $\mathbf{d}_1$ vanishes there, exactly where anatomical splay is ligament-locked to zero, so fading φ with the projection magnitude is both numerically and anatomically correct. All angles are clamped to anatomical ranges (`FLEX_LIMITS`).

3.3 Thumb chain

The thumb's carpometacarpal joint is a 2-DOF saddle; one palm-referenced angle cannot represent opposition. The decomposition therefore keeps the metacarpal *direction* $\hat{\mathbf{t}} \propto \mathbf{p}_2 - \mathbf{p}_1$ as a unit vector in palm coordinates (2 DOF, filtered on the sphere by filtering components and renormalizing), plus the CMC offset $\mathbf{p}_1 - \mathbf{p}_0$, and measures the two flexions about the thumb-plane normal

$$\hat{\mathbf{a}}_t \propto \hat{\mathbf{t}} \times (\mathbf{p}_3 - \mathbf{p}_2) \quad (\text{fallbacks } \hat{\mathbf{t}} \times (\mathbf{p}_4 - \mathbf{p}_3),\ \hat{\mathbf{z}} \times \hat{\mathbf{t}} \text{ when degenerate}), \qquad \theta_{\text{mp}},\ \theta_{\text{ip}} \tag{8}$$

as signed angles about $\hat{\mathbf{a}}_t$ down the chain. Constructing the axis from the same cross product makes $\theta_{\text{mp}} \geq 0$ by construction and θ_{ip} consistently signed in the thumb plane; when the thumb is straight the axis is ill-defined but both angles are ≈ 0, so the fallback chain is benign.

3.4 Re-synthesis and exactness

`synthesizeHand` is the forward kinematics of the model: knuckles at calibrated palm coordinates, phalanges chained by $R(\hat{\mathbf{a}}, \theta)$ with calibrated lengths, the thumb from its stored direction/axis. For in-range angles the pair (`decomposeHand`, `synthesizeHand`) is an exact inverse; the unit tests verify a 10^{-6}-radian round trip against an *independent* generator FK (`experiments/retarget_corpus.mjs`, a different parameterization), plus rotation invariance and left/right mirror symmetry.

3.5 Angle-domain filtering

Previously each landmark *position* was one-euro filtered [1]. Position filtering is shape-distorting: during a fast curl the three coordinates of a fingertip lag by different amounts (the derivative term reacts per-axis), so the transient shape is skewed and the pitch lags. The tracker instead filters each decomposed *angle* with a one-euro filter tuned $f_{c,\min} = 0.8\,\text{Hz}$, $\beta = 1.2$: at rest the effective cutoff is lower than the legacy position filter (less standstill jitter), and during a fast curl ($\dot{\theta} \approx 8\,\text{rad/s}$) the cutoff opens to $\approx 10\,\text{Hz}$ (near-zero lag). Measured on a scripted 0.25 s curl with the degradation model of §6.1: 50%-rise lag 86 ms (legacy) $\rightarrow$ 19 ms (tracker), Fig. 3.

3.6 Geometry calibration and pinch preservation

Knuckle palm-coordinates and bone lengths are EMA-calibrated with rate $\alpha = \max(1/(n{+}1), 0.02)$ and a per-update length-change clamp. A true pinch must survive re-synthesis *in meters* (the server's grasp curve and double-pinch clutch read the thumb–index gap): inside 50 mm the thumb tip (and half the IP joint) is slid along the gap direction so the synthesized gap approaches the measured one as the pinch closes (`preservePinch`), with weight $w = 1 - g_{\text{raw}}/0.05$.

4 Depth restoration

Monocular hand landmarkers regress depth with far weaker supervision than the image-plane coordinates; the dominant error is a per-hand multiplicative attenuation $\hat{z} = \kappa z$, $\kappa < 1$ — a finger curling toward the camera reads much flatter than it is. This is the primary cause of "the fingers don't curl."

4.1 Global attenuation estimate

Bone lengths are constant. For bone i with measured components $(\hat{b}_x, \hat{b}_y, \hat{b}_z)$ and calibrated length L_i, the rigid constraint $b_x^2 + b_y^2 + (\hat{b}_z/\kappa)^2 = L_i^2$ gives a per-bone estimate

$$\kappa_i \;=\; \frac{|\hat{b}_z|}{\sqrt{L_i^2 - \hat{b}_x^2 - \hat{b}_y^2}}, \tag{9}$$

well-conditioned exactly for *z-dominant* bones. The tracker aggregates $\hat{\kappa} = \sum w_i \kappa_i / \sum w_i$ with sensitivity weights $w_i = L_i |\hat{b}_z|$ over bones passing identifiability gates ($|\hat{b}_z| > \max(0.35L, 8\,\text{mm})$ and $L^2 - \hat{b}_{xy}^2 > (0.3L)^2$), EMA-filters it with a confidence-scaled rate, and rescales all landmark z by $1/\hat{\kappa}$ (the wrist is the origin, so the scaling is anchor-free). This corrects *every* bone — including mid-chain bones whose own length deficit is below the noise floor.

4.2 Per-bone residual restoration and its noise gain

After the global correction, each bone still short of its calibrated length can be restored individually: $|b_z| \leftarrow \sqrt{L^2 - b_x^2 - b_y^2}$, keeping the measured sign. The estimator's noise gain diverges near the image plane: $\partial|b_z|/\partial L = L/|b_z| \to \infty$ as $b_z \to 0$, so an unconditioned restoration hallucinates centimeters of depth with a random sign out of millimeter noise (observed: open-hand angle RMSE tripled). The correction is therefore weighted by the *significance* of the length deficit $\delta_i = (L_i - \|\mathbf{b}_i\|)/L_i$:

$$w_i = \mathrm{clip}\Big(\frac{\delta_i - \ell(L_i)}{0.13}, 0, 1\Big), \qquad \ell(L) = \max\big(0.12,\ 2.5\,\sigma/L\big), \quad \sigma = 4\,\mathrm{mm}, \tag{10}$$

so short (distal) bones, whose relative noise is large, restore only under extreme unambiguous attenuation.

4.3 Anthropometric anchor (breaking the circularity)

Both (9) and the residual restoration need *true* lengths, but a hand that starts curled calibrates its lengths from already-attenuated frames — the correction would then be circular. The palm bones (wrist→knuckle) lie near the image plane whenever the operator faces the camera, so *they* calibrate true; finger and thumb bones get an anthropometric floor proportional to the calibrated palm length L_p (wrist→middle knuckle), $L_i \geq 0.85\,\rho_i L_p$, with ratios ρ_i following standard hand-proportion data [6] (e.g. middle proximal/middle/distal = 0.466/0.296/0.222; `BONE_PRIOR`). The asymmetric max-tracking EMA (rise $\alpha_\uparrow = 0.06$, decay $\alpha_\downarrow = 0.02$) then learns each true length whenever the bone sweeps near the image plane, without noise ratcheting. The floor biases $\hat{\kappa}$ conservatively toward 1 (under-correction); the per-bone stage absorbs the residual.

4.4 Measured effect

On the degraded corpus ($\kappa = 0.65$, $\sigma = 2.5\,\mathrm{mm}$, §6.1), the full-fist angle RMSE drops from 13.4/13.6° (L/R, legacy) to 8.0/9.5° while the open hand stays comparable (2.7/2.3° → 3.5/5.3°); the tracker unit test recovers $\hat{\kappa}$ and bounds the tracked MCP error at < 0.17 rad against a ~ 0.3 rad uncorrected bias.

5 Two-hand and whole-body consistency

5.1 One similarity transform preserves contact exactly

Let $\mathbf{w}_h$ be the true wrist and δ_h the true wrist→tip offset of hand $h \in \{L, R\}$. The wire tip position is $\mathcal{S}(\mathbf{w}_h) + s_{\text{shape}}\, M\delta_h$. If the operator's tips touch, $\mathbf{w}_L + \delta_L = \mathbf{w}_R + \delta_R$, and the wire gap is

$$\big\|\mathcal{S}(\mathbf{w}_R) + s_{\text{shape}} M\delta_R - \mathcal{S}(\mathbf{w}_L) - s_{\text{shape}} M\delta_L\big\| \;=\; |s_{\text{shape}} - s|\,\|\delta_R - \delta_L\|. \tag{11}$$

The legacy pipeline had $s_{\text{shape}} = 1 \neq s$: with $s = 0.905$ (a 0.42 m-shouldered operator) and hands extended toward each other ($\|\delta_R - \delta_L\| \approx 0.3\,\mathrm{m}$) this alone is $\approx 29\,\mathrm{mm}$ of guaranteed gap error — exactly the 24 mm to 29 mm measured (Table 1). Setting $s_{\text{shape}} = s$ (option `scaleShape`) makes contact preservation *exact*: the measured wire gap error on ideal landmarks is 0.0 mm for every touch shape. Downstream, the server maps both hands' contact points through one fixed world transform with one reach scale (`engine._solve_side`), so coincident wire points remain coincident robot targets.

5.2 Shared-image-frame wrist-pair fusion

Pose-model world wrists carry centimeter-level *relative* error between the two hands, which (11) cannot fix. Both hands' normalized image landmarks, however, live in one image frame with pixel accuracy. With aspect-corrected image coordinates $\tilde{\mathbf{u}} = (u\,W/H,\ v)$, each hand's meters-per-normalized-unit scale is estimated from its knuckle span, $s_h = \|\mathbf{p}_5 - \mathbf{p}_{17}\|_{\text{world}} / \|\tilde{\mathbf{u}}_5 - \tilde{\mathbf{u}}_{17}\|$, and the image-plane inter-wrist vector $\bar{s}\,(\tilde{\mathbf{u}}_0^R - \tilde{\mathbf{u}}_0^L)$, $\bar{s} = \frac{1}{2}(s_L + s_R)$, replaces the pose-derived (x, y) components of $\mathbf{w}_R - \mathbf{w}_L$; the pair midpoint and the depth split are kept from the pose (the image cannot observe depth). Gates: the correction is rejected if s_L, s_R disagree by $> 1.8\times$ (hands at very different depths) and faded to zero as the image-implied separation grows from 0.30 m to 0.50 m — the span scale is estimated at *knuckle* depth, giving a perspective bias of $\sim$4% of the separation, negligible in the contact regime and harmful only for wide poses (measured: 7 mm residual on hands-on-hips at 0.30 m separation, zero effect on salute at 0.66 m where the gate disables fusion). Under the degradation model's ± 2 cm per-side pose bias the fusion removes the image-plane error; the remaining gap error equals the pose *depth* bias ($\Delta z \approx 33$ mm injected), a documented limitation (§8).

5.3 Elbows for arm posture

The arm IK's cost (see `mabel_controller.tex` §4) contains an elbow-direction cosine ($W_{\text{elbow}} = 30$) that was permanently disabled because ARKit headsets stream no elbows. The webcam *does* track elbows; the client now attaches `leftElbow/rightElbow` transforms mapped by $\mathcal{S}$, `mabel/wire.py` parses them into the existing `OperatorPose.{left,right}_elbow` fields, and `engine._solve_side` maps them through the same world transform as the contact point. Measured offline on the engine itself (§6.4): elbow-direction error drops by 15–40° on the iconic shapes at a 13 mm to 30 mm contact-tracking cost — the trade encoded by the stack's validated weights.

5.4 The palm-mirror convention

Feeding the finger solver true right-hand skeletons *extended* curling fingers (a pinch drove the index to its extension limit; measured thumb–index tip gap 66 mm for a 5 mm pinch). Root cause, established with the dorsal-normal lemma of §3.1: in the simulated ORCA model, closing flexion moves the right hand's fingertips toward $+\hat{\mathbf{f}} \times (\mathbf{p}_{\text{pinky}} - \mathbf{p}_{\text{index}})$ (measured cos $= +0.87$; left hand -0.87), whereas every anatomically right-handed skeleton curls toward the *opposite* side. The reference synthetic hands the stack was validated with (`vision_pro/docs/tools/exp_hand_retarget.synth_hand`, used by the end-to-end mock client) are built with the same inverted convention — the shipped stack is self-consistent but expects *mirrored* skeletons.

The client compensates without touching the validated server: `reflectPalm` reflects the hand across its own palm plane $\{\mathbf{p} : \hat{\mathbf{z}} \cdot (\mathbf{p} - \mathbf{p}_0) = 0\}$ and then translates so the *middle fingertip is pinned*:

$$\mathbf{p}' = \mathbf{p} - 2\,\hat{\mathbf{z}}\,\hat{\mathbf{z}}^{\top}(\mathbf{p} - \mathbf{p}_0) + \mathbf{c}, \qquad \mathbf{c} \text{ s.t. } \mathbf{p}'_{12} = \mathbf{p}_{12}. \tag{12}$$

A reflection is an isometry, so *all distances* (the pinch gap, the clutch thresholds) are preserved; the palm frame R_{op} is built from in-plane points and is unchanged; the arm's contact target (the middle tip) is pinned, so absolute tracking and the two-hand contact argument of (11) are unaffected; only the finger-shape reading flips to the convention the solver expects. With the reflection the offline solver closes the pinch to 12.8 mm (from 66 mm) and the fist MCP solves positive ($+0.62$ vs. -0.28 rad). If the simulated hand's convention is ever fixed at the source (MJCF regeneration), `mirrorPalm` is a single option to drop.

6 Experiments

6.1 Scripted landmark corpus and degradation model

`experiments/retarget_corpus.mjs` generates synthetic MediaPipe outputs for twelve canonical shapes — open palm, fist, four per-finger half-curls, thumb–index pinch (generated tip gap 4.9 mm), a two-hand fingertip *heart* (index tips meet exactly by construction; thumb pair 20.4 mm), hands-forward with index tips touching, salute, arms-up, hands-on-hips — each with exact ground-truth joint angles and operator-space landmark positions. The hand FK is an independent implementation from the client's, so round trips are genuine. The *degraded* variant models MediaPipe behavior: hand-landmark depth attenuation $\kappa = 0.65$ plus $\sigma = 2.5$ mm noise, per-side pose wrist/elbow biases of $\sim$2 cm to 3 cm plus noise, image-landmark noise $\sigma = 0.002$ (normalized), all with a fixed seed. Shoulder width is 0.42 m so the anthropometric scale ($s = 0.905$) exercises the scale-consistency path.

Replay is two-level: (1) *client-level* — `experiments/retarget_dump.mjs` runs the corpus through the actual exported synthesis functions in both pipelines (75 frames at 30 Hz per shape; metrics averaged over the last 15) and measures wire-frame angle errors, contact-point error, and fingertip-pair gaps; (2) *robot-level* — `experiments/exp_retarget.py` streams the dumped frames to the LIVE bridge (`ws://127.0.0.1:9090/teleop`, 4.5 s per shape after a sim reset), samples `robot_state`, captures simview screenshots *during* streaming, and measures robot fingertip positions by forward kinematics on an offline in-process model of the same scene (`mj_forward` only).

6.2 Client-level results

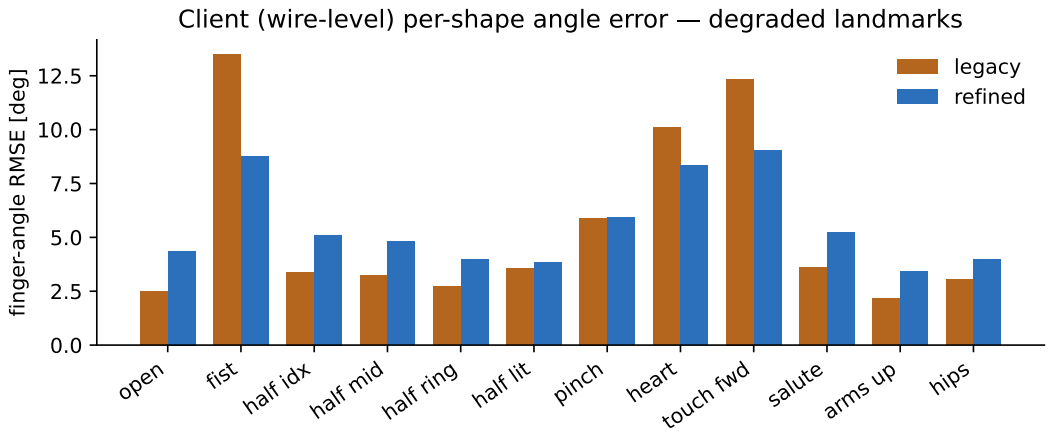


Figure 1: Wire-level per-shape finger-angle RMSE (mean of both hands) under the MediaPipe degradation model. The refined pipeline halves the fist error (depth restoration) while remaining comparable on already-flat shapes; on ideal landmarks (not shown) the refined pipeline is exact and the legacy pipeline errs only through its scale mixing.

6.3 Robot-level results (live stack)

Robot-level fingertip gaps carry a server-side floor independent of the client: the arm tracks only the *middle-tip* contact point (a fixed 0.14 m palm-forward offset on the robot side, regardless of curl),

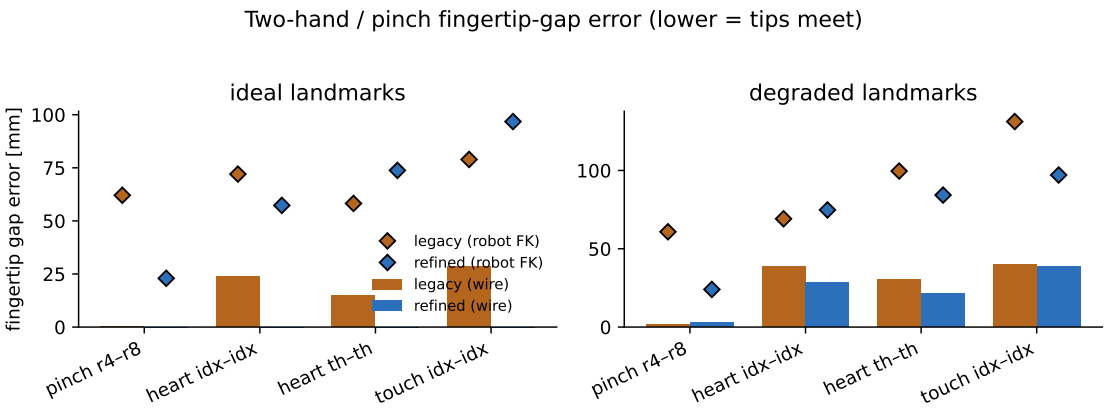


Figure 2: Fingertip-pair gap error for the touch shapes (bars: wire level; diamonds: robot level via FK against the reach-scaled operator gap). Ideal landmarks: the refined pipeline is exact at the wire. Degraded: the residual wire error is the pose-model *depth* bias the image fusion cannot observe.

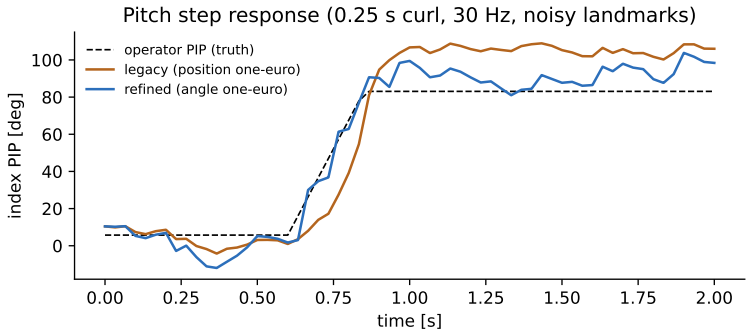


Figure 3: Pitch step response on a scripted 0.25 s index curl with degraded landmarks: angle-domain filtering (50% lag 19 ms) versus the legacy position-domain one-euro (86 ms).

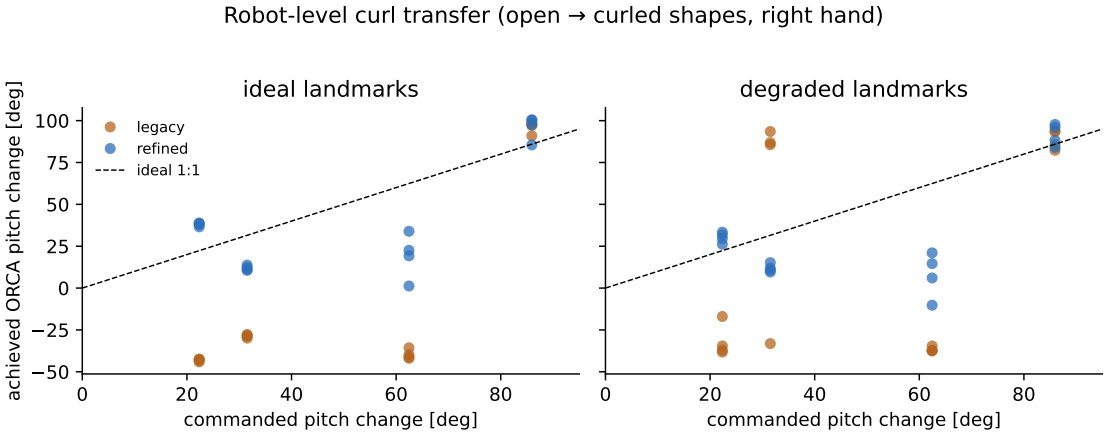


Figure 4: Curl transfer at the robot: achieved ORCA joint change from the open-palm baseline versus the commanded operator change, for every commanded MCP/PIP of the curled shapes (right hand). Points on the dashed line are perfect transfer.

the finger solver approximates shapes on a differently-proportioned hand, and the arm regularizers trade tracking for posture — together ∼55 mm to 95 mm of pair-gap even for *exact* wire input. Within that floor the refined client fixes the intra-hand thumb–finger coordination decisively — the pinch fingertip gap drops 67 mm→28 mm (ideal) and 66 mm→29 mm (degraded) — reduces the two-hand *touch-forward* gap 131 mm→97 mm (degraded), and restores correct curl *direction* (Fig. 4, Fig. 5): with the mirror compensation the pinch and fists close instead of hyper-extending (fist index MCP error −78.5° → −30.8°). The heart pair remains at the floor (69 mm→75 mm, within run-to-run spread). Screenshots of every shape (wide and hand close-up, captured *while* streaming) are archived under `experiments/retarget_runs/`.

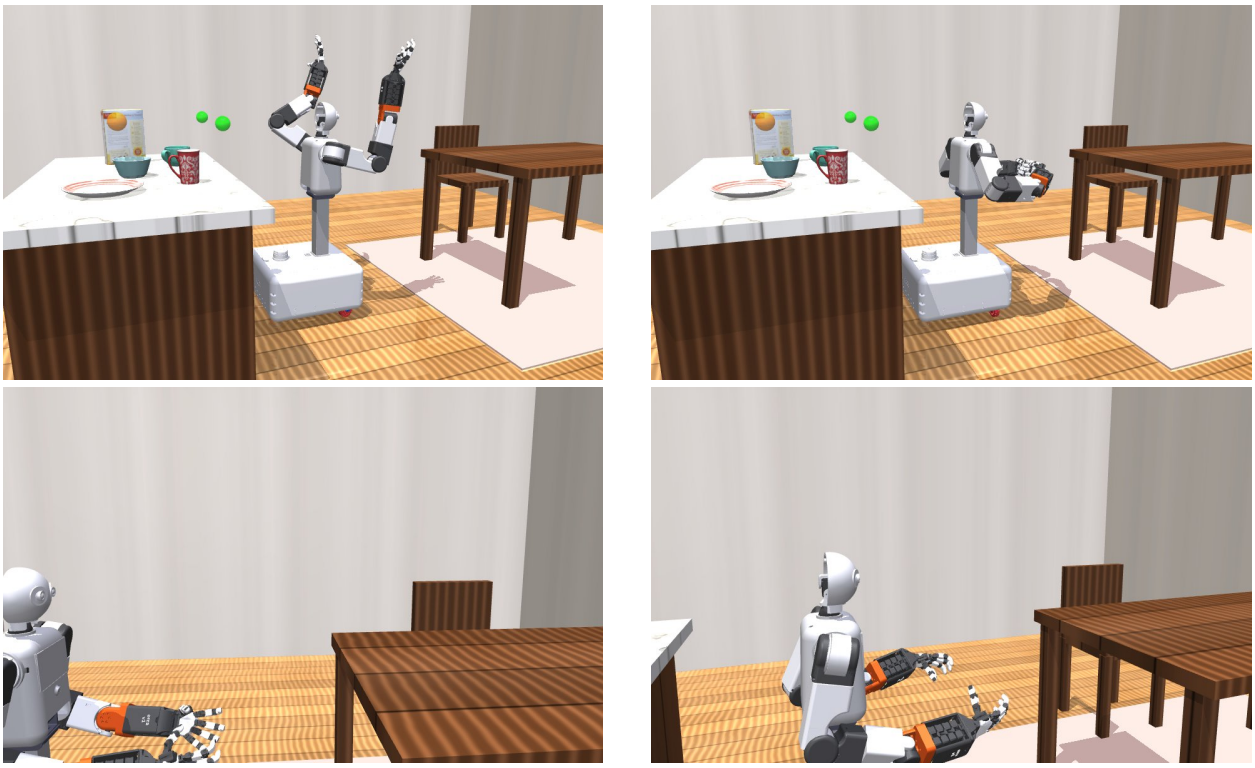


Figure 5: Live-stack screenshots captured during streaming. Top: refined pipeline on *arms up* (both arms overhead, open palms) and the two-hand *heart* (hands joined at the chest). Bottom: the *fist* hand close-up, legacy (left — fingers splayed open: the chirality inversion extends a curling hand) versus refined (right — fingers curled).

6.4 Elbow-direction study (offline engine)

6.5 Summary

7 Implementation and usage

Client. All math is exported pure from `web/js/bodyteleop.js`: `decomposeHand`, `synthesizeHand`, `refineHandShape`, `restoreDepth`, `estimateZScale`, `HandShapeTracker`, `reflectPalm`, `fuseWristPair`, `preservePinch`, plus the original frame functions. The browser runtime passes {`trackers, scaleShape:true, fuse:true, elbows:true, mirrorPalm:true`} to `synthesizeFrame`; omitting the options reproduces the legacy behavior exactly (used by the

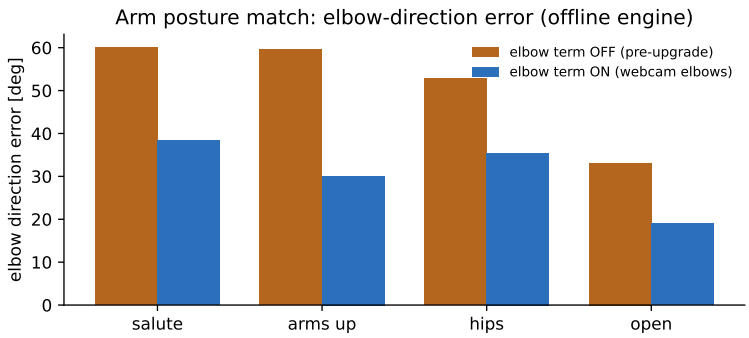


Figure 6: Elbow-direction error of the engine's arm solution with the elbow term off (pre-upgrade wire) vs. on (webcam elbows plumbed through), mean of both arms, 90 frames per shape, self-collision off to isolate the term.

Table 1: Per-shape before/after summary. aRMS = wire-level finger-angle RMSE (mean L/R, deg) on ideal/degraded landmarks; gap = first touch-pair fingertip gap error (mm) at the wire and at the robot (FK), degraded variant. Robot-level numbers include the server-side floor discussed in §6.3.

	aRMS ideal		aRMS degraded		gap err degraded (wire / robot)	
shape	legacy	refined	legacy	refined	legacy	refined
open palm	0.0	0.0	2.5	4.4	—	—
fist	0.0	0.0	13.5	8.7	—	—
half curls (4)	0.0	0.0	2.7–3.6	3.8–5.1	—	—
pinch	0.0	2.7	5.9	5.9	1.5 / 60.9	2.9 / 24.0
heart (idx–idx)	1.4	1.4	10.1	8.3	38.9 / 69.1	28.3 / 74.8
touch forward	0.0	0.0	12.3	9.0	40.0 / 131.2	38.6 / 97.1
salute / arms up / hips	0.0	0.0	2.1–3.6	3.4–5.2	—	—

harness as the "before" arm). The dexterity panel's `retargetLandmarks` (`web/js/dexterity.js`) now reuses `decomposeHand` on metric world landmarks and maps radians through the calibrated ORCA ranges (`handmodel.radToPct`).

Server (minimal). `controller/mabel/wire.py` parses optional `leftElbow`/`rightElbow`; `scripts/teleop/engine.py` maps a present elbow into the arm cost; `scripts/teleop/safety.py` fixes an ndarray-truthiness crash the elbows would have triggered. Nothing else server-side changed.

Reproduce. From `simulation_control_center/`:

```
node --test tests/js/*.test.mjs                                  # 95 tests
node experiments/retarget_dump.mjs --pipeline refined --variant degraded \
     --out /tmp/scratch/refined_degraded.json                    # x4 combos
python3 experiments/exp_retarget.py --dump .../refined_degraded.json  # live
python3 experiments/exp_retarget_offline.py --dump .../refined_ideal.json
python3 experiments/retarget_figures.py --data .../retarget # figures+table
```

8 Limitations

- **Inter-hand depth bias.** The image fusion corrects only the image-plane inter-wrist vector; a systematic pose-model depth bias between the wrists passes through ($\approx$ 33 mm in the degradation model). A span-ratio depth estimate is possible in principle but its per-frame noise ($\sim$5% of camera distance) exceeds the bias; left as future work.
- **Conservative $\hat{\kappa}$.** The anthropometric length floor biases the global attenuation estimate toward 1 (measured $\hat{\kappa} \approx 0.86$ for true 0.65 from a cold start in a fist); the per-bone stage compensates, at reduced correction for mid-chain bones.
- **Server-side gap floor.** Two-hand fingertip meeting at the robot is limited by the engine tracking only the middle-tip contact and by ORCA-vs-human finger proportions ($\sim$50 mm to 80 mm even for exact input). Closing this requires server-side work (e.g. tracking the touching fingertip pair), out of scope here.
- **Simulated hand chirality.** The mirror compensation (§5.4) matches the model as it exists; whether the *real* ORCA right hand shares the convention must be verified on hardware before using this path on the real robot, and the MJCF finding should be resolved at the source.
- **Pinch thresholds under scaling.** The wire pinch gap scales with s; for extreme operators (s near the 0.5/3.0 clamps) the fixed server thresholds (30 mm clutch-on) shift accordingly.
- **Abduction under mirror.** The reflection flips splay signs; the solver consumes mirrored splay consistently with the mirrored model, but the wire skeleton's abduction no longer reads anatomically (visible only to debugging tools that decompose the wire).

References

[1] G. Casiez, N. Roussel, D. Vogel. "1-Euro Filter: A Simple Speed-based Low-pass Filter for Noisy Input in Interactive Systems." *CHI*, 2012.

[2] F. Zhang et al. "MediaPipe Hands: On-device Real-time Hand Tracking." *CVPR Workshops*, 2020.

[3] V. Bazarevsky et al. "BlazePose: On-device Real-time Body Pose Tracking." *CVPR Workshops*, 2020.

[4] A. Handa et al. "DexPilot: Vision-Based Teleoperation of Dexterous Robotic Hand-Arm System." *ICRA*, 2020.

[5] Y. Qin et al. "AnyTeleop: A General Vision-Based Dexterous Robot Arm-Hand Teleoperation System." *RSS*, 2023.

[6] A. Buryanov, V. Kotiuk. "Proportions of Hand Segments." *Int. J. Morphology* 28(3), 2010.

[7] S. Yang et al. "ACE: A Cross-platform Visual-Exoskeletons System for Low-Cost Dexterous Teleoperation." arXiv:2408.11805, 2024.

[8] Y. Park, P. Agrawal. "Using Apple Vision Pro to Train and Control Robots." 2024.